Target Triage with Protein Language Models: Pancreatic Ductal Adenocarcinoma

1. Approach and re-ranking

In module 2, PDAC genes were ranked using RNA expression, protein abundance, and somatic mutation evidence. Here, a sequence based prior was added to ask whether there targets resemble proteins with clinical drug-target precedence. The purpose was to prioritize candidates for further assessment. Top 15 targets from the targets_pdac.csv was selected, and their Ensemble IDs were then converted to gene symbols, and was matched to proteome.tsv while preserving the embedding row positions. All 15 targets were mapped, the data pack contained 20,190 proteome proteins and a 1,201 protein training panel, each represented by 960-dimensional ESM C 300M embedding. The leakage guard was applied before training, it removed zero panel proteins because none of the selected targets overlapped the panel. The guard was still applied, and the empty removal list is saved as removed_training_targets.csv. Table 1 is the hand-off ranking exported to targets_pdac_w3.csv (w=0.6).

Rank	Gene	Module 2 rank	p_sm	p_ab	final
1	CEL	4	0.716265	0.009346	0.88
2	MUC16	1	0.00008	0.150975	0.76
3	LRP1B	5	0.284008	0.472467	0.71
4	COL11A1	3	0.099664	0.000297	0.65
5	BRSK2	6	0.266869	0.000118	0.64
6	XDH	9	0.503499	0.003453	0.63
7	CFTR	2	0.044407	0.001009	0.61
8	ACADL	7	0.152717	0.00002	0.55
9	NOS1	8	0.209032	0.000706	0.53
10	NPTX2	11	0.054379	0.500657	0.52

Table 1: Nearest drugged neighbors for every target are in the exported CSV

Did the prior agree with module 2? Largely no. Across the 15 targets, the Spearman correlation between the module 2 score and the best modality tractability prior is -0.14 (p=0.62). The two lines of evidence are close to independent. That is what one wants from a new line of evidence, but it also means the prior reshuffles the list instead of confirming it.

The target that moved most was CFTR, which fell from rank 2 to 7. Both its priors are close to zero, the second-lowest tractability of the 15. Its nearest drugged neighbors (GUCY2C, SCN7A, CASR; cosine 0.95) are multi-pass membrane proteins, so the embedding does place CFTR among membrane drug targets. Even so, the classifier scored it as untractable. CFTR is the target of approved small-molecule modulators (ivacaftor), but because CFTR is

not in the training panel, the prior never saw that label. Those drugs bind unusual sites within the transmembrane domains, and those sites do not resemble the classic pockets of the panel's enzyme.

The biggest upward moves (+3) were CEL (from 4 to 1), XDH (from 9 to 6), and FN1 (from 15 to 12). CEL and XDH are enzymes with the two highest small-molecule priors (0.716 and 0.503). XDH serves as a useful sanity check since it's the validated target of allopurinol and febuxostat, and the prior correctly identifies it as small-molecule-tractable without knowing its label. I'm more skeptical about FN1's p_sm value of 0.482. Fibronectin is a large extracellular matrix protein, and its clinical applications are antibody-based (specifically antibody fusions against its EDB domain), not small-molecule-based.

There are three reasons as to why w=0.6 was chosen, first, the module 2 score is the disease-specific evidence, while the prior knows nothing about PDAC, so the disease evidence should carry more weight. Second, the prior's family-split AUROC is 0.73, it is informative but far from decisive. Third, the module 2 scores of the top 15 are compressed into a narrow range (0.932-0.979), so converting them to percentiles exaggerates tiny differences. Giving the prior 40% lets it break these near ties without overriding the disease biology. A sensitivity check supports this choice. At w=0.8, the top two revert toward module 2 (MUC16, CEL, CFTR). CFTR is the most weight-sensitive target (rank 3 at w=0.8, rank 12 at w=0.4), which confirms that the prior's main effect on this list is its judgement of CFTR.

2. **Modality: a target where p_sm and p_ab disagree**

CEL showed the strongest preference for small molecules, with p_sm=0.716 versus p_ab=0.009. This is biologically reasonable because CEL is a lipid-digesting enzyme with a catalytic site and substrate-binding pocket that small molecules could potentially target. Although CEL is secreted, this does not automatically make it suitable for antibody therapy. In comparison, MUC5B favors antibodies and is a large extracellular mucin rather than an enzyme with a defined catalytic pocket. Together, these results suggest that protein structure and function, alongside localization, help explain the modality difference.

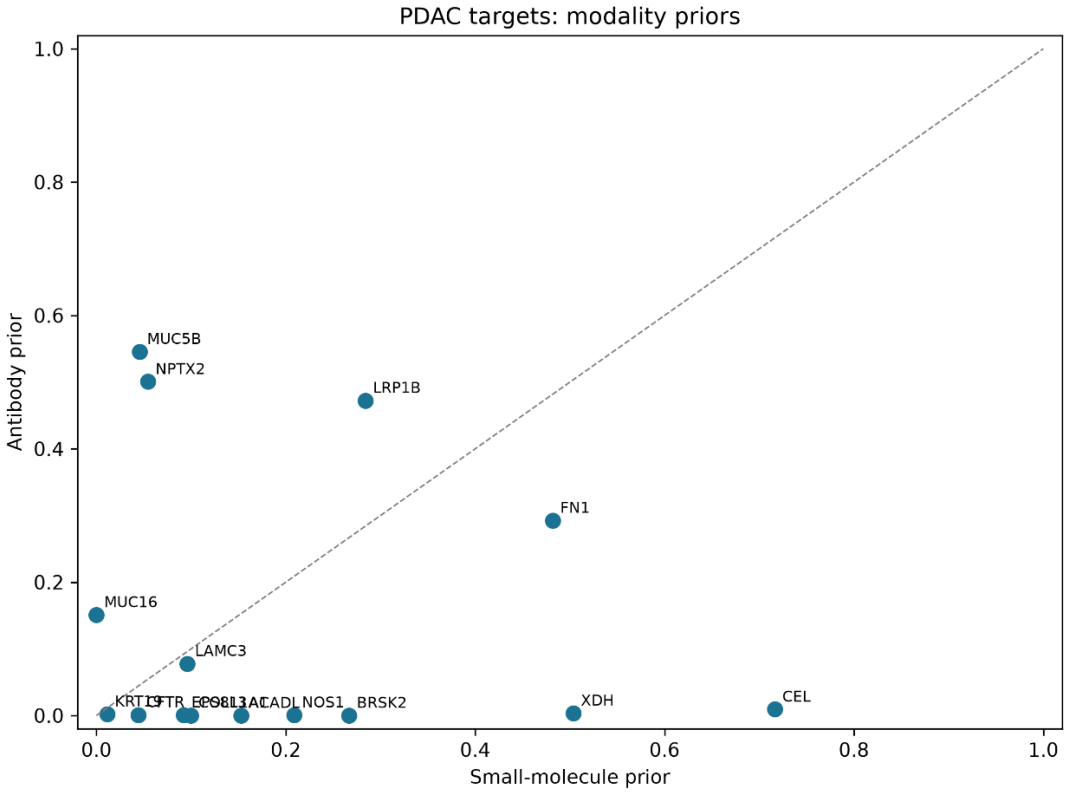


Figure 1 Modality map of the 15 PDAC targets

However, CEL's nearest clinical-precedence neighbors GPA33, MMP9, and TNKS2, have different functions, so their embedding similarities do not prove a shared binding mechanism. MUC16's low antibody prior despite existing antibody-based approaches also illustrates the model's limitations, sequence truncation is one possible explanation. Finally, predicted tractability does not establish therapeutic benefit. CEL was downregulated at both RNA and protein levels in the module 2 analysis, which could reflect changes in pancreatic tissue composition rather than a cancer dependency. Structural and functional studies would therefore be needed before deciding whether targeting CEL would help treat PDAC.

3. Validation, why the family split matters

Split	AUROC	Average precision
Random split (StratifiedKFold)	0.848	0.727
Family split (GroupKFold on family_group)	0.732	0.548
Chance level (prevalence of positives)	0.500	0.334

Table 2 Five-fold cross-validated performance of the any-clinical prior on the training panel (n=1201)

The family-split values, AUROC 0.732 and AP 0.548, are the numbers to trust. In a random split, close paralogs from the same protein family land on both sides of the split. Their embeddings are nearly identical, so the model gets credit for recognizing family membership

rather than for learning transferable features of tractability. Holding out whole families tests what was actually needed, whether the prior generalizes to proteins whose family it has not been labeled. The targets include several families with little or no drugging history, such as mucins, keratins and EPS8-like proteins, so this is a relevant setting. The random split would have hidden a drop of 0.116 in AUROC and 0.179 in AP, overstating the prior's reliability. Still, an AUROC of 0.3 means it often misranks pairs, and small differences between targets should not be overinterpreted. One limitation is that the validation was run on the any-clinical label, and the separate small-molecule and antibody priors were not validated separately.

4. Anchor variants: Sequence Scores versus disease effects

Gene	Variant	Score	Within-protein percentile
APOE	R176C (epsilon2)	-8.559	11.8
APOE	C130R (epsilon4)	7.208	100.0
TREM2	R47H	-5.156	47.7
TP53	R175H	-5.974	19.3

Table3. Variant effect scores (Uniprot numbering, which includes APOE's 18-residue signal peptide)

The anchor variants show that sequence-model scores do not directly predict disease risk. APOE R176C (ε2) scored -8.559, although ε2 generally reduces Alzheimer's risk, while C130R (ε4) scored +7.208 despite its association with increased risk. These scores describe how well a substitution fits learned sequence patterns, a negative score means less expected, while a positive score means more favored. Therefore, an evolutionary plausible substitution can still have harmful effects in humans. APOE positions use precursor numbering, including its 18-residue signal peptide.

TREM2 R47H scored -5.156 at the 47.7th percentile, while TP53 R175H scored -5.974 at the 19.3rd percentile. Their established disease relevance shows that a variant does not need an extreme sequence score to matter biologically. These percentiles indicate ranking within each protein's score distribution, not the probability of disease. Because the scores alone cannot establish gain or loss of function, interactions, or tissue-specific effects, they need to be interpreted alongside functional, structural, and clinical evidence.

5. Limitations and Targets for Week 4

Sequence embeddings provide useful clues about tractability, but they cannot confirm an accessible drug-binding pocket or establish whter targeting a protein would benefit PDAC. The predictions may reflect historical research bias and incomplete sequences, especially for large proteins. Module 2 also used gene-level integration with separate normal cohorts, so strong signals could reflect tissue-composition changes rather than tumor dependencies.

For week 4, CEL and XDH will be prioritized for structure-based pocket assessment. CEL ranked first overall (0.880, p_sm=0.716), making its catalytic pocket a clear hypothesis to investigate through pocket geometry, accessibility, and local structural confidence. XDH ranked sixth, offering a second enzyme candidate for assessing catalytic pockets an the influence of cofactors and domain arrangement. Both were downregulated in module 2, so finding a suitable pocket would support structural tractability, which functional experiments would still be needed to determine whether inhibition is appropriate.

6. References and Analysis Sources

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AI assistance: Anthropic. (2026). Claude Opus 5.5 [Large language model]. https://claude.ai was used for editing and formatting only, the analysis and content were all mine.